

Auditing Sparse Autoencoder Feature Edits in GPT-2 Small: Coordinates, Fidelity, and a Lexical Pilot

Anonymous Authors¹

Abstract

Steering with a sparse autoencoder (SAE) edits an activation along a feature direction and reports the edit as a change of that feature. For GPT-2 small and a public layer-8 residual SAE we check that report in three steps and propose no method. (a) *Readout*. Because GPT-2’s LayerNorms ignore the all-ones direction, an edit should be read canonically, as $E(P(h + \delta))$. Read raw, as $E(x + \delta)$, a model-invisible edit appears to move the SAE by 124.7 times its norm; canonically the change is at most 3.8×10^{-15} . (b) *Same-layer fidelity*. In a conditional pilot (8 features, 54 held-out documents) with the decoder, encoder-row and corrected edits matched to the target at every test point, projected-decoder edits leave non-target drift of 0.656α (active targets, dose q_{99} , medians), and a least-norm correction posed with $J_E(Ph)P$ leaves 0.114α . In dense toy regimes the same correction moves drift onto newly activated features. (c) *Lexical behavior*. In a pre-registered lexical pilot on 32 held-out articles (PPLM’s space word list; a lexical proxy, not semantic control), no SAE-feature edit at a KL cap of 0.1 nats changed space-lexicon occurrence relative to no edit. The correction and the decoder agreed on every article (0 of 32 discordant; under an unverified independence model, the upper end of a two-sided 95% Clopper–Pearson interval for the discordance probability is 10.9%). Lower internal drift did not translate into a lexical difference at this budget; the pilot cannot show equivalence, and larger budgets were not run.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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1. Introduction

Sparse autoencoders (SAEs) are used to steer language models by editing an internal activation along a learned feature, typically by adding a multiple of the feature’s decoder vector or by clamping the feature inside the SAE reconstruction (Templeton et al., 2024; O’Brien et al., 2025; Wu et al., 2025). The edit is then reported in the SAE’s vocabulary: “feature j was increased by α ”. For one public model–SAE pair, GPT-2 small with a residual SAE at layer 8, we check that sentence in a fixed order:

- Readout*. Which input does the SAE read after an edit (Section 4)?
- Same-layer fidelity*. Among edits the model can see, how much of the re-encoded change lands on the target, on a common denominator of test points (Section 5)?
- Lexical behavior*. Does a cleaner internal edit change a lexical target more at matched output change (Section 6)?

The central question is narrow: for this GPT-2 SAE, how do internal fidelity in valid coordinates and a limited lexical effect differ? We answer it for one feature, one budget and one lexical target, and freeze the study at that scope.

Much of the machinery is known. Encoder and decoder directions differ (Chanin et al., 2025; O’Neill et al., 2025), and misalignment has been named as one cause of steering failure (Luo et al., 2026). SAE-targeted steering fits an effect map and reports that its pseudo-inverse did worse (Chalnev et al., 2024). Least-norm readout edits and null-space protection are standard (Geiger et al., 2024; Wu et al., 2024; Fang et al., 2025; Cui et al., 2026), and LayerNorm’s invariance to the mean direction is elementary. We propose no method and claim three bounded results:

- Readout convention* (Section 4). The canonical readout $E(P(h + \delta))$, not the raw $E(x + \delta)$, is the quantity that corresponds to what the model receives. The raw convention turns a model-invisible edit into an apparent $124.7\times$ feature change. This is a measurement caveat, not a new property of LayerNorm.

- *Conditional same-layer fidelity* (Section 5). With decoder, encoder row and correction all matched at every test point, the correction lowers same-layer non-target drift relative to the projected decoder in a 8-feature pilot. Controlled toy regimes show where it instead moves drift onto newly activated features.
- *A null lexical pilot* (Section 6). At a KL cap of 0.1 nats, no SAE-feature edit changed space-lexicon occurrence on 32 held-out articles, and the correction and the decoder never disagreed. The pilot is small and its events are sparse, so this is a null at one budget, not equivalence.

We make no claim about semantic steering, and lower internal drift or KL is never taken as evidence that a method is better.

2. Related Work

SAE steering and its evaluation. Feature steering by clamping or adding decoder directions was demonstrated at scale (Templeton et al., 2024) and applied to refusal (O’Brien et al., 2025). AxBench steers with $h + \alpha w_{\text{dec}}$, scores concept, instruction relevance and fluency with a judge, and reports a mean steering score of 0.165 for SAEs against 0.239 for difference-in-means (Wu et al., 2025); we therefore keep a difference-in-means baseline. Input features and output features rarely coincide (Arad et al., 2025). Other work edits in the SAE latent space and decodes with the error term added back (Bayat et al., 2025), fits decoder-space edits to counterfactual activations (Makelov et al., 2025), or scores encoder-decoder interventions by output effects (Bhalla et al., 2025). These works measure behavior; they do not re-encode the edited activation.

Targeting SAE features. SAE-TS learns a linear map $\hat{y} = xM + b$ from steering vectors to the SAE activations of steered rollouts and steers with $M_j / \|M_j\| - \lambda Mb / \|Mb\|$; its appendix reports that the pseudo-inverse of M is worse on all nine tasks (Chalnev et al., 2024). Feature-guided activation additions reuse this map (Soo et al., 2025). We instead re-encode each edited activation at the edit layer with the exact per-input encoder rows and control the edit norm. The pseudo-inverse result is prior evidence against our correction.

Encoders are not inverses of decoders. Absorption leaves holes in encoder recall (Chanin et al., 2025), a linear-nonlinear encoder has an amortisation gap (O’Neill et al., 2025), and encoding a bare steering vector is dominated by the encoder bias (Mayne et al., 2024). Luo et al. (2026) conclude that off-manifold artifacts, “not just encoder-decoder misalignment”, contribute to steering failures. Side effects

have been predicted from decoder geometry and logit effects (Duan, 2026; Khan et al., 2026); we measure the target’s own same-layer readout instead.

Least-norm and null-space interventions. Distributed interchange interventions (Geiger et al., 2024) and LoReFT (Wu et al., 2024) are least-norm edits that set orthonormal readouts to target values; our correction has the same form with non-orthogonal SAE encoder rows. AlphaEdit and AlphaSteer project updates onto null spaces of preserved keys or benign activations (Fang et al., 2025; Sheng et al., 2026); LEACE and MiMiC are least-change edits (Belrose et al., 2023; Singh et al., 2024). Cui et al. (2026) hold SAE features fixed with within- and across-layer null-space projectors while changing behavior, which is our protected-set constraint used adversarially. Readout and intervention directions are related through a metric (Park et al., 2024), so the Euclidean norm is one choice among several.

Benchmarks. SAEBench (Karvonen et al., 2025) and RAVEL (Huang et al., 2024; Chaudhary & Geiger, 2024) evaluate interventions downstream; on our reading, none compares intended with re-encoded change. Jacobian SAEs (Farnik et al., 2025) reduce, with an identity in place of the MLP, to the $E_{\text{active}} D_{\text{active}}$ matrix of our linear analysis.

3. Problem Setup

SAE. Let $x \in \mathbb{R}^d$ be an activation, written in the coordinates the SAE was trained on. The encoder has pre-activations $p(x) = Ex + b$ with $E \in \mathbb{R}^{m \times d}$ and features $a = \sigma(p)$, where σ is ReLU, JumpReLU with threshold θ , or TopK. The decoder $D \in \mathbb{R}^{d \times m}$ has columns D_j . For the public SAE in Section 4, $p(x) = W_{\text{enc}}^\top (x - b_{\text{dec}}) + b_{\text{enc}}$, so $E = W_{\text{enc}}^\top$ and $b = b_{\text{enc}} - W_{\text{enc}}^\top b_{\text{dec}}$.

Edit problem. An edit is specified by a target j , an intended post-activation change $\alpha > 0$, and a protected set Q , which we take to be the non-target features active at x . The target value $a_j + \alpha$ requires the pre-activation change $\Delta p_j^{\text{req}} = a_j + \alpha - p_j$ whenever the target ends up active. The achieved change of an edit δ is $\Delta a = \sigma(p(x + \delta)) - \sigma(p(x))$.

Definition 3.1 (Edit-fidelity decomposition). With active sets A_0 and A_1 before and after the edit, the *target error* is $|\Delta a_j - \alpha|/\alpha$, and the drift terms are ℓ_2 norms of Δa over non-target features $i \neq j$ in $A_0 \cap A_1$ (D_{kept}), $A_0 \setminus A_1$ (D_{deact}) and $A_1 \setminus A_0$ (D_{new}), with $D_{\text{orig}} = (D_{\text{kept}}^2 + D_{\text{deact}}^2)^{1/2}$ and D_{all} over all $i \neq j$. Terms are reported relative to α (target-matched) or to the norm budget ρ (equal norm).

For ReLU, JumpReLU and TopK a feature inactive before and after does not change, so $D_{\text{all}}^2 = D_{\text{orig}}^2 + D_{\text{new}}^2$.

Edits compared. The decoder edit $\delta = cD_j$; the rescaled decoder, with c set so that $\Delta p_j = \Delta p_j^{\text{req}}$ (per input in the toy study; the deployable weight-only $\alpha/(E_j D_j)$ in the real protocol, which is not matched); the encoder row $\delta \propto E_j^\top$; the LN correction $\delta = M^+ t$ with $M = E_{S \cup Q}$ (rows for $S = \{j\}$ and Q) and $t = (\Delta p_j^{\text{req}}, 0, \dots, 0)$; in the toy study also bounded repair, a trust region $\arg \min_{\|\delta\| \leq \tau} \|M\delta - t\|$, and random directions.

Admissible edits and readouts. For GPT-2, adding $c\mathbf{1}$ to the residual at the hook leaves the logits unchanged (Proposition 4.1), so a model-visible edit is the class $\delta + \text{span}(\mathbf{1})$. We represent it by $P\delta$ with $P = I - \mathbf{1}\mathbf{1}^\top/d$, the member in $\text{range}(P)$, where the SAE’s inputs $x = Ph$ lie. This is a choice of representative, not the only valid space. We read an edit *canonically*, $\Delta a = \sigma(p(P(h + \delta))) - \sigma(p(Ph))$, which depends only on the class. Every actor edit below lies in $\text{range}(P)$, so for it the raw and canonical readouts coincide. The LN correction linearizes the canonical readout, whose Jacobian is $J_{\text{eff}} = J_E(Ph)P$; it therefore solves $(MP)\delta = t$. A *mean-only* edit along $\mathbf{1}$ is a control, never a method.

Comparison modes and units. In an *equal-norm* comparison, every direction is scaled to the same $\|\delta\|$. In a *target-matched* comparison, every direction is scaled so that $\Delta a_j = \alpha$. Rows where no non-negative scale achieves this are kept as infeasible. The independent unit is a random toy instance in Section 5.3 and a held-out document in Section 5.2. Features, positions and base points within a unit are averaged before any summary.

4. Raw and Canonical Coordinates

We executed a model–SAE contract before measuring anything: GPT-2 small and the residual SAE at `blocks.8.hook_resid_pre` ($d = 768$, $m = 24576$) at pinned revisions, read through the canonical TransformerLens/SAELens path and through the plain Hugging Face model our adapter uses, on the same token ids (Section E).

Proposition 4.1 (Mean invariance, GPT-2 only). *Consider the GPT-2 architecture in evaluation mode and exact arithmetic: each block reads the residual stream only through `ln_1` and `ln_2`, the unembedding reads it only through `ln_f`, and each LayerNorm subtracts the mean over the d model coordinates and divides by the standard deviation of the centred vector. Adding $c_t\mathbf{1}$ to the input of block ℓ at any set of positions t leaves every logit unchanged.*

This is a standard consequence of mean subtraction in LayerNorm, and the same property that lets TransformerLens centre writing weights; we state it only to fix its scope. We make no claim for other architectures: RMSNorm, for example, does not subtract the mean. The proof is in Section B.

Readouts. The SAE was trained on $x = Ph$, where h is the Hugging Face hidden state. The canonical readout $E(P(h + \delta)) - E(Ph)$ depends only on the class $\delta + \text{span}(\mathbf{1})$. The raw off-slice readout $E(x + \delta) - E(x)$ feeds the SAE an input outside $\text{range}(P)$ whenever $\mathbf{1}^\top \delta \neq 0$, and then depends on a component the model ignores. Linearizing the canonical readout gives $J_{\text{eff}} = J_E(Ph)P$ with $J_{\text{eff}}\mathbf{1} = 0$; the least-norm solution of $(MP)\delta = t$ lies in $\text{range}(PM^\top) \subseteq \text{range}(P)$.

What the checks show (Table 1).

- *Implementation parity.* In float64 our fast path matches the canonical path to 10^{-13} – 10^{-15} for activations, features and post-edit distributions. The pre-registered float32 check fails its probability tolerance through rounding, amplified by a common logit shift of up to 279 from `center_unembed`; we keep that failure and run everything in float64. Uncentred inputs give FVE 0.003 against 0.83.
- *Two readouts, two quantities.* A mean-only edit of norm 10.1 leaves the model unchanged (KL 1.1×10^{-15}). Read raw, as the contract’s K9 check did, it moves the SAE by 124.7 times its norm; read canonically, by at most 3.8×10^{-15} . We first described the raw number as internal change without external effect. That reading is withdrawn: it is an off-training-subspace stress test of the encoder, not a change of the model’s features. For unprojected decoder rows, whose mean component is only 3–6% of their norm, the two readouts differ substantially while the model’s outputs agree to rounding. For the 760 admissible edits at 16 pilot test points they agree to 3.6×10^{-12} , so no admissible result depends on the convention.

5. Admissible Feature Realization

This section asks how much of a model-visible edit’s re-encoded change lands on the target and where the rest goes: first in the real model, on a common denominator of test points, and then in controlled toy regimes that show when a local correction redistributes drift.

5.1. What linear algebra already fixes

Several standard facts decide which outcomes are forced (statements in Section A, proofs in Section B); in the real model E stands for EP , the Jacobian of the canonical readout. Rescaling fixes the target coordinate but removes leakage on Q only if column j of ED restricted to $S \cup Q$ is a multiple of e_j (Proposition A.1). A least-norm correction $M^+ t$ exists exactly when $t \perp \ker M^\top$; otherwise the residual certifies infeasibility (Proposition A.2). For ReLU,

Table 1. Raw off-slice versus canonical SAE readouts of the same edits, and what the model sees (float64). Contract windows: the 8 stored K9 cases, recomputed. Pilot points: 16 test points of the REAL-01R pilot, all stored values reproduced exactly. The full contract table is Table 11.

Edit / check	Quantity	Raw $E(x+\delta)$	Canonical $E(P(h+\delta))$	Model
mean-only, contract windows (8)	drift/ ρ (median)	124.7	3.8×10^{-15} (max)	KL 1.1×10^{-15}
mean-only, pilot points (16)	D_{all}/ρ	99 (median)	1.4×10^{-13} (max)	KL 1.4×10^{-15}
unprojected decoder row	$ \Delta D_{\text{all}} /\rho$, raw vs canon.		up to 0.34	$ \Delta \log p $ 2.8×10^{-13}
admissible edits (760 incl. controls)	raw – canonical		at most 3.6×10^{-12}	–
fast vs canonical, float64	features / post-edit KL	1.6×10^{-14}	1.5×10^{-15}	pass
fast vs canonical path, float32	next-token $ \Delta p $ (tol. 10^{-5})		2.8×10^{-5}	FAIL (kept)

holding protected pre-activations fixed leaves only newly activated drift, and a post-activation Jacobian cannot activate an inactive target (Proposition A.5); JumpReLU and TopK add range gaps and evictions (Proposition A.6); norm budgets give a ridge solution (Proposition A.7).

5.2. Real-model pilot

Protocol (P3-REAL-01R pilot). Fixed before any test document was read: calibration and test sets are WikiText-103 validation and test articles; 64 features are drawn from a firing-density band on calibration windows, and the pilot uses a seeded subsample of 8; per-feature target changes are the calibration activation quantiles q_{50}, q_{90}, q_{99} and $2q_{99}$, and equal-norm budgets are fractions of the median residual norm, all frozen with a hash. Each feature has up to 8 test documents with an active-target point and 8 with an inactive-target point. All edits are admissible and are compared at equal norm or at matched target change; unmatched rows are kept. Quality is $\text{KL}(P_{\text{clean}} \| P_{\text{edit}})$ and ΔNLL over the 16 positions after the edit, for 2 documents per feature and target type. The unit is the test document, with 95% document-cluster bootstrap intervals. The run (float64, two CPU threads) produced 6144 rows from 54 documents; the median LN solve took 22 ms.

Results (Table 2). For active targets the target-matched decoder edit leaves $D_{\text{orig}} = 0.583\alpha$ and $D_{\text{all}} = 0.656\alpha$; the weight-only rescaled decoder is identical because pre-activations are affine. The LN correction removes D_{orig} by construction and leaves only newly activated drift, $D_{\text{all}} = 0.114\alpha$, with a smaller edit norm (12.7 against 21.6). Per document (LN minus decoder, active targets, q_{99}), $\Delta D_{\text{all}} = -0.503 [-0.584, -0.421]$, $\Delta D_{\text{new}} = -0.17 [-0.202, -0.142]$ and $\Delta \text{KL} = -0.00377 [-0.0117, -0.00025]$ nats; the signs hold at every dose (Table 16, Figure 2). For inactive targets ΔD_{all} and ΔKL are negative at every dose, but the ΔD_{new} intervals include zero at q_{50} and q_{90} . Against the encoder row, the paired difference in D_{all} (LN minus encoder row) is $-0.687 [-0.822, -0.613]$ at nearly the same norm.

Denominators. Re-aggregating the stored rows with a target tolerance of 10^{-6} fixed before the tables were produced shows that decoder, encoder row and LN are matched at every test point and dose (largest error 5.1×10^{-8} , float32 rounding of the stored reference), so their comparison uses the full common denominator. Random directions are infeasible at 52%–55% of points and are never ranked. One earlier descriptive gate readout (G1) pooled matched active-target rows (0.583 [0.452, 0.6]) with unmatched inactive-target rows (0.342 [0.289, 0.371]); its side is unchanged. The gate’s minimum-unit rule and a target-gain metric were added while the pilot’s test run was executing and are labelled in-run amendments. Each pilot feature’s q_{99} rests on 26–134 positive calibration activations, so for 3 features it is effectively a top order statistic; we keep the frozen doses. These are internal measurements; Section 6 explains why they do not show external benefit.

5.3. When the correction redistributes drift

A correction that protects some features can only move drift elsewhere: onto unprotected features in a linear model, or onto features that cross threshold in a ReLU model (Proposition A.5). Whether it also *reduces* total drift depends on the regime, which controlled toy studies isolate. These results are unfavorable to the correction in the dense regime and are reported here, not in the appendix.

Controlled toy study: design. The toy study (Python standard library) fixed its hypotheses, metrics, unit (one random draw of (D, E, b)), budget and stop conditions in a configuration before the reported run; development seeds 0–2 were used only for debugging, and seeds 1000–1019 were run once (20 instances per cell). One change followed the development seeds: an exploratory sparsity axis (Section C). All 260/260 feasibility checks and verification identities passed (Table 5); they show that the code is correct and are not findings. Every number below is re-aggregated from the raw files.

Linear regimes (toy). Seven linear regimes fall into the classes that Propositions A.1 and A.2 predict in all 20 in-

Table 2. Target-matched fidelity with its full denominator (REAL-01R pilot, dose q_{99} , float64; medians over test documents). “Matched”: test points whose target error is at most 10^{-6} , out of all test points; drift for a row with fewer matched points is over its feasible or deployed rows only, and the inactive rescaled row misses the target (error 0.727) and is not comparable. Drift is relative to α ; KL is over the 16 positions after the edit (2 documents per feature and target type). Mean-only rows exist only under the raw readout and are omitted. Intervals: Table 14; status counts: Table 13.

Target	Method	matched	n_{doc}	D_{orig}/α	D_{new}/α	D_{all}/α	$\text{KL} \times 10^3$	$\ \delta\ $
active	decoder	64/64	42	0.583	0.311	0.656	5.32	21.6
	rescaled (weight-only)	64/64	42	0.583	0.311	0.656	5.32	21.6
	encoder row	64/64	42	0.681	0.366	0.778	1.03	11.8
	LN ($J_E P$)	64/64	42	≈ 0	0.114	0.114	0.96	12.7
	random	31/64	24	7.33	108	108	691	438
inactive	decoder	64/64	40	0.583	0.361	0.716	3.41	36.9
	rescaled (weight-only)	0/64	40	0.342	0.112	0.37	0.65	22.5
	encoder row	64/64	40	0.943	0.943	1.36	1.38	21
	LN ($J_E P$)	64/64	40	≈ 0	0.295	0.295	1.16	22.5
	random	29/64	23	17.4	347	348	327	1174

stances (Table 4). Rescaling fixes pure scale errors but not interference; the LN correction removes interference at a larger norm; protecting all non-targets with $|S \cup Q| > d$ is infeasible, and protecting a subset can *raise* total leakage (R4b: 1.73 to 2.16).

Dense active sets (toy, pre-registered). With $d = 16$, $m = 64$ and a median of 13 active ReLU features, the LN correction removes most originally active drift, but its larger norm pushes inactive features over threshold. Its total drift beat the rescaled decoder’s in only 10/20, 3/20 and 0/20 instances for active targets at $\alpha = 0.25, 1, 4$ (Table 9). Bounded repair reached crossing-free edits only by abandoning the target (median target error 0.23–0.84), and a post-activation Jacobian never moved an inactive target (Proposition A.5).

Sparser active sets (toy, exploratory). An axis added after inspecting development seeds lowers the median active count to 1 (ReLU bias -0.75). There the LN correction had lower total drift in 19/20, 18/20 and 17/20 instances (Figure 1); at bias -0.5 it won at $\alpha \leq 1$ but only 4/20 at $\alpha = 4$. Our reading, a hypothesis and not a finding: a local correction helps only when the protected set is small relative to d and few inactive features cross threshold.

Where the real SAE falls. In the pilot, the SAE has about 72.4 active features per token out of $d = 768$, a protected fraction far below that of the dense toy. Its behavior resembles the sparse exploratory settings rather than the dense pre-registered one.

- The LN correction’s newly activated drift did not exceed the decoder’s: for active targets $\Delta D_{\text{new}} = -0.17$ $[-0.202, -0.142]$ at q_{99} (Table 16).
- Its total drift was lower at every dose.

This fits the regime hypothesis stated after the exploratory toy study. It is a pilot with 8 features and one layer, however, so it is consistent with that hypothesis rather than a test of it. The full calibration set of 64 features and other layers or SAE families would be needed before any regime claim; they were not evaluated (Section 7).

6. Lexical Change and Quality

Everything above is measured with the encoder that defines the edit. This section states what that can certify and then reports one limited behavioral pilot.

6.1. What agreement with the encoder can certify

Where it is feasible, the LN correction makes $D_{\text{orig}} = 0$ and the target error zero by construction, as measured with the same encoder. That says nothing about features outside $S \cup Q$ (Proposition A.5) or about any other readout. A synthetic example makes this concrete. A ground-truth dictionary A ($d = 16$, $m = 8$, pairwise cosine 0.3) defines an external readout $z = A^+ x$, and the SAE is a noisy estimate of A (Table 10, Section D). The LN correction drives the internal error below 10^{-6} in every case, yet the external error falls from 0.221 to $< 10^{-6}$ when the encoder is exact (better in 20/20 instances), rises from $< 10^{-6}$ to 0.239 when the decoder is exact (better in 0/20), and is unchanged when both are noisy (better in 11/20). These outcomes are fixed by construction: the internal metric measures agreement with the encoder. Behavior can also change while SAE readouts are held fixed (Cui et al., 2026).

6.2. Space-lexicon pilot

Design. One behavioral pilot was frozen before any corpus count (Section H). The metric is *space-lexicon occurrence*: whether 32 greedy tokens contain a word from the 18-word space list of PPLM (Dathathri et al., 2020), matched as

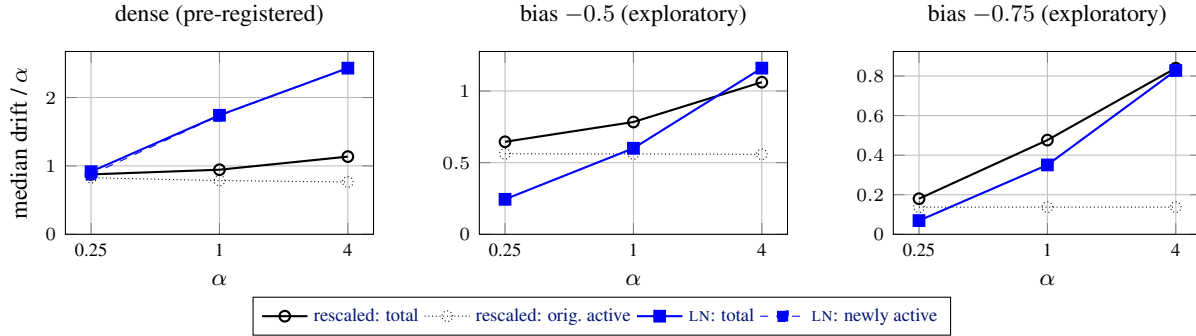


Figure 1. Where drift goes (ReLU, active targets, 20 instances, medians). In the dense setting the LN correction trades originally active drift for more newly activated drift. In the sparser exploratory settings it lowers total drift at small α . The LN “newly active” series coincides with its “total” series, because the correction leaves no originally-active drift.

whole lowercased words. The topic was fixed in advance, not chosen by comparing topics; an earlier wedding lexicon was closed because its calibration support was too small. The same lexicon selects the SAE feature and the DiffMean direction, so this is a supervised lexical objective and not an independent semantic check. Support was 58 of 1943 calibration windows in 22 of 60 articles (minimum 20 windows), and 28 of those windows contain the polysemous “star”. Every actor, feature 4671, received the same KL-capped budget of $0.1 \times$ the median norm ($\kappa = 0.1$ nats). The unit is the test article; the primary estimand, fixed in advance, is the mean over articles of any-hit(LN) minus any-hit(decoder) at those budgets; every other comparison and metric is secondary and descriptive. All 32 test articles attempted gave a prompt, and no row failed. 28 of them had served as edit sites in the internal pilot, so they are held out from calibration but not unseen by this project.

Result (Table 3). Without an edit, 1 of 32 continuations contains a list word: “All-Star” matches “star” under the frozen whole-word rule, a legitimate hit of a polysemous entry that we neither removed nor filtered afterwards. The decoder, encoder-row and LN edits gave exactly that outcome on every article. The primary difference, LN minus decoder in any-hit, is therefore 0.0 percentage points with 0 discordant articles (0 LN-only, 0 decoder-only). Its document-bootstrap interval is $[0.0, 0.0]$ because every observed per-article difference is zero; a degenerate interval is not evidence of equivalence. As a model-based sensitivity statement only: if the 32 articles were independent draws of a discordance indicator, zero discordances would bound the discordance probability by 8.9% (one-sided 95%) or 10.9% (upper end of a two-sided 95% Clopper–Pearson interval), and since $|\mathbb{E}[\text{hit}_{\text{LN}} - \text{hit}_{\text{dec}}]| \leq \Pr(\text{discordance})$ under the same model, the same numbers bound the mean difference. That model is not verified: the articles form one fixed seeded sequence, 28 of them were edit sites in the internal pilot, and one prompt per article was used, so these are not population

estimates. The pre-registered claim that LN exceeds the decoder is not supported by this pilot. DiffMean differed from no edit on 3 articles (gained 2, lost 1; net +1), a secondary and descriptive comparison (Table 18). The edits were not inert. Descriptively and post hoc, the greedy continuation changed in 24–26 of 32 articles for the SAE-feature edits and in 31 for DiffMean, a text-change count that is not a substitute outcome. The canonical target change at prompt positions was 1.73 (decoder), 3.92 (encoder row) and 2.99 (LN); LN had the lowest non-target drift (1.68 against 3.65) and the lowest realized test KL (0.0205 against 0.0281), below the calibration cap $\kappa = 0.1$ that selected the budget. None of this moved lexicon occurrence. The mean-only control reproduced every unedited continuation (32/32). At this one KL cap, feature edits changed internal readouts and surface text but not space-lexicon occurrence, and the pilot cannot separate LN from the decoder. It says nothing about larger budgets, other features, other SAEs or other behaviors, and the generated text was not evaluated semantically (Section 7).

7. Limitations

- *One model, one SAE, one layer.* The fidelity results come from a pilot with 8 features on 54 documents and are conditional on that sample. WikiText differs from the SAE’s OpenWebText training data, and quality metrics come from 2 documents per feature and target type.
- *The invariance proof is GPT-2 specific.* Architectures whose residual readers are not mean-subtracting need their own analysis.
- *float32 is characterized, not fixed.* The float32 contract failed its pre-registered tolerance. We ran everything in float64.
- *One unrecorded detail.* The TransformerLens version used to train the SAE is not recorded; our reconstruc-

Table 3. Space-lexicon pilot (32 held-out WikiText-103 articles, one prompt each, 32 greedy tokens, edit at every position, float64). Budget: calibration-selected multiple of the median centred norm $\bar{r} = 101.6$ under $\kappa = 0.1$ nats. KL is teacher-forced on the common prompt; NLL is that of the continuation under the unedited model; Δa_j is the canonical target change at prompt positions. Any hit: mean over articles with a 95% document-bootstrap interval. A lexical proxy, not semantic success.

Method	n	budget	cal. KL	test KL	any hit (%)	hits/100 tok.	NLL	Δa_j
no edit	32	—	—	0	3 [0, 9]	0.0977	1.32	0
decoder PD_j	32	$0.1\bar{r}$	0.0299	0.0281	3 [0, 9]	0.0977	1.2	1.73
encoder row $PE_j^\top$	32	$0.1\bar{r}$	0.0303	0.0269	3 [0, 9]	0.0977	1.27	3.92
LN ($J_E P$)	32	$0.1\bar{r}$	0.0205	0.0205	3 [0, 9]	0.0977	1.37	2.99
DiffMean Pv	32	$0.1\bar{r}$	0.0539	0.0487	6 [0, 16]	0.391	1.34	0
mean-only (control)	32	$0.8\bar{r}$	—	1.5×10^{-17}	3 [0, 9]	0.0977	1.32	0

tion check (L0 72.4, FVE 0.83 against the release’s 60 and 0.9) supports the coordinates but cannot pin down the training code.

- *Toy results are small and largely forced*; the sparse axis is exploratory.
- *Internal numbers are internal*. KL and Δ NLL measure total output change, not success.
- *The behavioral pilot is small and lexical*. It has 32 articles and one feature, and hits are rare (1 of 32 without an edit, from a polysemous word), so it cannot detect modest differences. The lexicon also chose the steered feature and the DiffMean direction. Semantic success is NOT RUN.
- *The dose was not extended*. The KL cap limited every actor to $0.1 \times$ the median norm, and the same topic was not rerun at larger caps. The null therefore holds for this budget only.
- *The sensitivity bounds are model-based*. 28 of the 32 test articles had been edit sites in the internal pilot, and the independence model behind the discordance bounds is unverified.
- *Not evaluated*. Two follow-ups lie outside the frozen scope of this study and were not run: the fidelity pilot over the full set of 64 calibration features (and other layers or SAE families), and an independent semantic evaluation of the generated text. The conditional fidelity claim and the lexical null do not depend on them; a regime claim or a semantic claim would.
- *Other choices*. The Euclidean norm is a modeling choice (Park et al., 2024). JumpReLU and TopK families (Lieberum et al., 2024) add the boundary effects of Proposition A.6 and are not covered.

8. Conclusion

For GPT-2 small and its layer-8 residual SAE, an edit’s feature change should be read canonically, because the raw

readout reports large changes for edits the model cannot see. With that fixed and edits restricted to model-visible directions, a least-norm correction lowers same-layer drift relative to the projected decoder in a conditional pilot, whereas in a dense toy regime it moves drift onto newly activated features. In a pre-registered lexical pilot, the lower internal drift did not change space-lexicon occurrence at a KL cap of 0.1 nats, and the correction could not be separated from the decoder. We freeze the study at this scope. Whether cleaner feature edits steer behavior, lexically at larger budgets or semantically, remains open.

Impact Statement

This work concerns the reliability of interpretability-based interventions, which are sometimes used to support safety claims such as removed capabilities or suppressed behaviors. Checking the readout convention and separating internal from external effects should reduce over-claiming. A raw readout can report large SAE changes for edits the model cannot see, which matters for SAE-based monitoring; null-space machinery that holds SAE readouts fixed also bears on SAE-based defenses (Cui et al., 2026). We do not develop attacks, and our behavioral task is benign topic steering.

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A. Standard Facts

These statements are standard or elementary; none is claimed as new.

Linear encoders.

Proposition A.1 (Mismatch and rescaling). *If σ is the identity, then $\Delta a = \alpha E D e_j$ for $\delta = \alpha D_j$. Rescaling hits the target if and only if $(ED)_{jj} \neq 0$, and then leaves leakage $\alpha \|(ED)_{Qj}\| / |(ED)_{jj}|$ on Q . It therefore suffices if and only if column j of ED , restricted to $S \cup Q$, is a multiple of e_j . For a tied encoder $E = D^\top$ whose decoder columns have pairwise cosine c , the leakage is $c\sqrt{m-1}\alpha$ both before and after rescaling.*

Proposition A.2 (Least-norm solutions and infeasibility). *$M\delta = t$ is solvable if and only if $t \perp \ker M^\top$. If M has full row rank, $\delta^* = M^\top (MM^\top)^{-1}t$ is the unique minimum-norm solution, and every other solution has squared norm $\|\delta^*\|^2 + \|n\|^2$ for some $n \in \ker M$. Otherwise the residual $r = t - MM^\top t$ certifies infeasibility: $M^\top r = 0$ and $r^\top t = \|r\|^2 > 0$. In particular, a generic target is infeasible once $|S \cup Q| > \text{rank } M$, which holds whenever $|S \cup Q| > d$. Two identical encoder rows with targets $(\alpha, 0)$ leave residual $\alpha/\sqrt{2}$ in any dimension.*

These are textbook facts (Golub & Van Loan, 2013; Björck, 1996).

Corollary A.3 (Three regimes). *Let Q contain all non-target features.*

- If $E = D^+$ and D has full column rank, then $\delta^* = \alpha D_j$: the decoder edit is already least-norm.
- If D has orthonormal columns and $E = \text{diag}(s)D^\top$, then δ^* is the rescaled decoder edit.
- If $E = D^\top$ with Gram matrix $G = D^\top D$, then $\|\delta^*\|/\alpha = \sqrt{(G^{-1})_{jj}} \geq 1$, with equality if and only if $Ge_j \propto e_j$.

Remark A.4 (Compute). For same-layer edits, the Jacobian of p restricted to $S \cup Q$ is the gathered matrix $E_{S \cup Q}$. With $|S \cup Q| \ll d$ it is small, so materializing it costs little. Matrix-free solvers that use only Jacobian–vector and vector–Jacobian products, such as CGLS/LSQR (Paige & Saunders, 1982; Björck, 1996), matter when the protected readouts sit at a later layer. In the toy study, CGLS matched the dense solution to within 7.5×10^{-13} (relative).

Nonlinear boundaries.

Proposition A.5 (ReLU piece exactness). *For ReLU, suppose no pre-activation, including the target’s, changes sign between x and $x + \delta$. Then $\Delta a = \text{diag}(\mathbb{I}[p > 0])E\delta$ exactly. If the pre-activations of $Q = A_0 \setminus \{j\}$ are held fixed, then $D_{\text{kept}} = D_{\text{deact}} = 0$, and all remaining drift is D_{new} . It comes from inactive features with $(E\delta)_i > -p_i$. The almost-everywhere Jacobian row of an inactive target is zero, so a solve on post-activation Jacobians cannot activate it.*

Proposition A.6 (JumpReLU and TopK). *A JumpReLU feature cannot take values in $(0, \theta]$, and crossing θ changes the activation by at least θ where the almost-everywhere derivative predicts no change. Under TopK with exactly k positive active features, an inactive target can enter only by evicting at least one of them. In both cases the local derivative is not a global inverse. For TopK, comparing active sets at the two endpoints only lower-bounds the number of changes along the path.*

Proposition A.7 (Norm budget). *$\min_{\|\delta\| \leq \tau} \|M\delta - t\|$ is convex. Its solution is M^+t if $\|M^+t\| \leq \tau$. Otherwise it is $(M^\top M + \lambda I)^{-1}M^\top t$ with the unique $\lambda > 0$ such that the solution has norm τ (Moré & Sorensen, 1983).*

B. Proofs

Proposition A.1. With σ the identity, $p(x + \delta) - p(x) = E\delta$. For $\delta = cD_j$ this is cED_j . Choosing $c = \alpha/(ED)_{jj}$ sets the target coordinate to α , and the Q coordinates are then $\alpha(ED)_{Qj}/(ED)_{jj}$. For $E = D^\top$ with unit columns and pairwise cosine c , $(ED)_{jj} = 1$ and $(ED)_{ij} = c$, which gives $c\sqrt{m-1}\alpha$.

Proposition A.2. The range of M is the orthogonal complement of $\ker M^\top$. Under full row rank, $MM^\top$ is invertible and $\delta^* = M^\top (MM^\top)^{-1}t$ lies in $\text{range } M^\top \perp \ker M$. Any other solution is $\delta^* + n$ with $n \in \ker M$, so the Pythagorean identity gives minimality. In general $MM^\top$ is the orthogonal projector onto $\text{range } M$. Hence $r = t - MM^\top t \in \ker M^\top$ and $r^\top t = r^\top (r + MM^\top t) = \|r\|^2$, which is nonzero iff $t \notin \text{range } M$. For duplicate rows e with targets $(\alpha, 0)$, both residual coordinates depend on the same scalar $u = e^\top \delta$. The minimum of $(u - \alpha)^2 + u^2$ is at $u = \alpha/2$ with value $\alpha^2/2$. The other rows can be satisfied simultaneously when e and those rows are linearly independent.

Corollary A.3. If D has full column rank, then $(D^+)^+ = D$. For $E = \text{diag}(s)D^\top$ with orthonormal D , $E^+ = D \text{diag}(s)^{-1}$. For $E = D^\top$, $(D^\top)^+ = DG^{-1}$, and $\|DG^{-1}e_j\|^2 = e_j^\top G^{-1}GG^{-1}e_j = (G^{-1})_{jj}$. By Cauchy–Schwarz in the G -inner product, $1 = (e_j^\top e_j)^2 \leq (e_j^\top G e_j)(e_j^\top G^{-1} e_j)$, and $G_{jj} = 1$.

Proposition A.5. Along $x + s\delta$ for $s \in [0, 1]$, each $p_i(s)$ is affine in s . Its sign therefore changes on the segment iff it differs at the endpoints. If no index changes activity, then σ acts as a fixed diagonal 0/1 map on the segment. Protected pre-activations are held fixed, so protected activations do not change. An inactive feature changes only if p_i becomes positive. For an inactive target, the j -th row of $\text{diag}(\mathbb{1}[p > 0])E$ is zero.

Proposition A.6. JumpReLU outputs $p \mathbb{1}[p > \theta]$, whose range is $\{0\} \cup (\theta, \infty)$. Crossing from $p \leq \theta$ to $p > \theta$ changes the output by more than θ . Under TopK, the active set has at most k elements. If k positive features are active and the target enters, one must leave. The k -th order statistic of affine functions is not monotone along a segment, so endpoint comparison is only a lower bound.

Proposition A.7. The objective is convex quadratic and the constraint set is convex. The KKT conditions are $(M^\top M + \lambda I)\delta = M^\top t$ with $\lambda \geq 0$ and $\lambda(\|\delta\| - \tau) = 0$. Convexity rules out the “hard case” of indefinite trust-region problems (Moré & Sorensen, 1983).

Proposition 4.1. Write $\text{LN}(v) = \gamma \odot (v - \bar{v}\mathbf{1})/\sqrt{s(v)^2 + \epsilon} + \beta$ with $\bar{v} = \mathbf{1}^\top v/d$ and $s(v)^2 = \|v - \bar{v}\mathbf{1}\|^2/d$. For any scalar c , $v + c\mathbf{1} - (\bar{v} + c)\mathbf{1} = v - \bar{v}\mathbf{1}$, so the centred vector, s , and hence LN are unchanged. Let $r_t^{(k)}$ be the residual entering block k at position t , and let the edited run add $c_t\mathbf{1}$ to $r_t^{(\ell)}$ for t in some set T ($c_t = 0$ elsewhere). We show by induction on $k \geq \ell$ that $\tilde{r}_t^{(k)} = r_t^{(k)} + c_t\mathbf{1}$ for every t . The base case is the edit. In GPT-2, block k computes $u_t = r_t + \text{Attn}(\text{LN}_1(r_1), \dots, \text{LN}_1(r_t))_t$ (causal attention with learned absolute positions, which enter only at the embedding) and $r_t^{(k+1)} = u_t + \text{MLP}(\text{LN}_2(u_t))$. By the hypothesis every LN_1 input differs by a multiple of $\mathbf{1}$, so all queries, keys, values and attention outputs are unchanged, and $\tilde{u}_t = u_t + c_t\mathbf{1}$. The same argument applied to LN_2 gives the claim for $k + 1$. After the last block, ln.f removes $c_t\mathbf{1}$, and the logits $W_U \text{ln.f}(\cdot)$ are unchanged. Dropout is inactive in evaluation mode. The argument uses exactly the listed structure: a residual reader that is not mean-subtracting (RMSNorm, a direct read of the residual, or a post-LN block) breaks it, so we make no claim beyond GPT-2. In floating point the statement holds up to rounding (Table 11, K9).

Representatives. The logits are a function of h modulo $\text{span}(\mathbf{1})$ at each edited position, so any linear complement of $\text{span}(\mathbf{1})$ represents model-visible edits. $\text{range}(P)$ is the orthogonal complement and the one where the SAE’s training inputs lie; choosing it is a convention. The canonical readout $E(P(h + \delta))$ is invariant to $\delta \mapsto \delta + c\mathbf{1}$ because $P\mathbf{1} = 0$; the raw readout $E(Ph + \delta)$ is not unless $E\mathbf{1} = 0$.

Effective Jacobian. With $p(x) = Ex + b$ and $a = \sigma(p)$, the canonical readout of δ is $\sigma(EP(h + \delta) + b)$; its pre-activation Jacobian in δ is EP and its almost-everywhere post-activation Jacobian is $\text{diag}(\sigma'(p(Ph)))EP = J_E(Ph)P$. For the LN correction, $(MP)^+t \in \text{range}((MP)^\top) = \text{range}(PM^\top) \subseteq \text{range}(P)$, and for $\delta \in \text{range}(P)$ we have $M\delta = MP\delta$, so the constraint is met exactly whenever it is feasible within $\text{range}(P)$.

C. Controlled Study Details

Pre-registration and deviations. The configuration fixed the following before the reported run: verification hypotheses V1–V8, metrics, units, seeds, a 120 s and two-thread budget, and stop conditions. V1–V5 and V7 are identities; failing any of them would indicate a bug. V6 and V8 are descriptive.

After running development seeds 0–2, we made changes before the reported run (v1 to v2):

- the CGLS tolerance of one feasibility check was made explicit (10^{-6});
- a low-rank-rows check was added;
- the ReLU bias axis (the exploratory sparser setting of Section 5.3) was added and marked exploratory.

The v1 sections were not modified.

A later bug in the summary code dropped Boolean flags from aggregates and reported a median-only external-improvement flag. We fixed it and reran. The raw files were bit-identical to the first run (SHA-256 values recorded in the internal run manifest; the raw files themselves ship with the supplement).

Regimes and methods.

- Linear regimes use $d = 16$ with $m = 8$ (equiangular, orthonormal) or $m = 48$ (Gaussian columns). Encoders are pseudo-inverse, tied, or scaled tied, with scales log-uniform in $[0.5, 2]$.
- The active-set setting uses $d = 16$, $m = 64$ and $E = \text{diag}(s)(\frac{1}{2}D^\top + \frac{1}{2}D^+)$ with s in $[0.7, 1.4]$. Base points are sums of three decoder atoms with coefficients in $[0.5, 1.5]$ plus noise 0.02.
- Activation settings: ReLU with bias -0.25 ; JumpReLU with $\theta = 0.3$; TopK with $k = 6$.
- Doses are $\alpha \in \{0.25, 1, 4\}$, and budget multipliers are $\{0.5, 1, 2\}$ times $\|\alpha D_j\|$.

Compute ledger.

- Development runs on seeds 0–2 took 8.4 s and 11.3 s.
- A component timing smoke took about 11 s.
- The reported run took 77.1 s; the rerun after the summary fix took 76.0 s.
- Regenerating the paper’s tables from the raw files takes about 1 s.

All of these ran single-threaded on a CPU.

D. Additional Tables

Table 4. Linear regimes (20 instances each; medians of per-instance means). Leakage is $\|\Delta a_{-j}\|/\alpha$. Norms (in parentheses) are relative to $\|\alpha D_j\|$. “Resid.” is the relative residual of the LN system.

Regime	Class	Dec. tgt err	Dec. leak	Resc. leak (norm)	LN leak (norm)	LN resid.
R0: $E=D^+$, $m < d$	no mismatch	$< 10^{-6}$	$< 10^{-6}$	$< 10^{-6}$ (1)	$< 10^{-6}$ (1)	$< 10^{-6}$
R1: orth. D , $E=\text{diag}(s)D^\top$	rescale suffices	0.365	$< 10^{-6}$	$< 10^{-6}$ (1.13)	$< 10^{-6}$ (1.13)	$< 10^{-6}$
R2: tied, $c=0.3$	removed by LN	$< 10^{-6}$	0.794	0.794 (1)	$< 10^{-6}$ (1.14)	$< 10^{-6}$
R3: scaled tied, $c=0.3$	removed by LN	0.34	0.884	0.989 (1.09)	$< 10^{-6}$ (1.24)	$< 10^{-6}$
R4a: tied, $m > d$, $P=\text{all}$	infeasible	$< 10^{-6}$	1.73	1.73 (1)	0.466 (0.398)	0.821
R4b: tied, $m > d$, $ P =7$	moved to unprot.	$< 10^{-6}$	1.73	1.73 (1)	2.16 (1.35)	$< 10^{-6}$
R5: $E=D^+$, $m > d$, $ P =7$	moved to unprot.	0.669	0.467	1.44 (3.1)	2.21 (3.6)	$< 10^{-6}$

Table 5. Feasibility and boundary checks (20 seeds; counts of passing rows).

Check	Pass
Full row rank: least-norm, minimality, CGLS = dense	80/80
More constraints than d : residual > 0 with certificate	40/40
Rank-deficient, target in range: residual ≈ 0	40/40
Low-rank rows ($k=8$, rank 5): certificate	20/20
Duplicate rows, targets $(\alpha, 0)$: residual $\alpha/\sqrt{2}$	20/20
ReLU: negative target unattainable	20/20
JumpReLU: target in $(0, \theta]$ unattainable	20/20
TopK: entering target evicts a feature	20/20

E. Model–SAE Contract

Table 11 lists the executed checks with their pre-registered tolerances and both verdicts, and Table 12 the contract fields, their sources and their status.

Table 6. Norm budgets (pre-registered dense ReLU, active target, $\alpha = 1$; medians over 20 instances). $\tau = \|\alpha D_j\|$.

Method	$\ \delta\ /\tau$	Target err	Total drift
decoder	1	0.339	0.597
rescaled decoder	1.57	$< 10^{-6}$	0.944
LN correction	3.17	0.00955	1.74
LN, scaled to $\ \alpha D_j\ $	1	0.629	0.278
trust region, 0.5τ	0.5	0.676	0.202
trust region, τ	1	0.4	0.361
trust region, 2τ	1.93	0.137	0.819
LN + repair	0.935	0.501	0.301

 Table 7. Pre-registered active-set setting for all activations (same columns as Table 9). $\dagger$: target value range-infeasible (JumpReLU, $\alpha < \theta$).

Act.	Target	α	Rescaled decoder			tgt err	LN correction			LN + repair	
			orig. act.	new	norm		orig. act.	new	norm	tgt err	total
ReLU	active	0.25	0.827	0.207	1.57	0.00955	0.0509	0.874	3.17	0.272	0.288
ReLU	active	1	0.786	0.51	1.57	0.00955	0.0509	1.73	3.17	0.501	0.301
ReLU	active	4	0.765	0.815	1.57	0.00955	0.043	2.43	3.17	0.558	0.338
ReLU	inactive	0.25	1.69	0.798	4.06	0.115	0.218	3.14	7.63	0.84	0.725
ReLU	inactive	1	0.862	0.803	2.26	0.0792	0.127	2.41	4.12	0.746	0.343
ReLU	inactive	4	0.603	1.07	1.82	0.0674	0.0931	2.38	3.51	0.653	0.307
JumpReLU	active	0.25	1.18	1.6	1.54	$< 10^{-6}$	$< 10^{-6}$	2.54	3.22	0.227	0.892
JumpReLU	active	1	0.928	1.2	1.54	$< 10^{-6}$	$< 10^{-6}$	2.63	3.22	0.487	0.701
JumpReLU	active	4	0.855	1.02	1.54	$< 10^{-6}$	$< 10^{-6}$	2.59	3.22	0.546	0.499
JumpReLU	inactive †	0.25	1.53	1.51	1.96	1	0.0306	2.66	3.54	1	0.903
JumpReLU	inactive	1	0.935	1.25	1.72	0.00982	0.0803	2.57	3.06	0.575	0.68
JumpReLU	inactive	4	0.611	1.25	1.65	0.0101	0.0512	2.42	3	0.597	0.473
TopK	active	0.25	1.39	0.993	1.59	$< 10^{-6}$	1.36	1.57	2.11	$< 10^{-6}$	0.206
TopK	active	1	0.942	0.837	1.59	$< 10^{-6}$	0.865	1.35	2.11	0.511	2.38
TopK	active	4	0.707	0.86	1.59	$< 10^{-6}$	0.364	1.32	2.11	0.394	0.794
TopK	inactive	0.25	1.36	1.08	1.66	1	1.08	1.4	2.07	1	0.603
TopK	inactive	1	0.98	0.878	1.59	$< 10^{-6}$	0.944	1.28	2.02	0.465	1.67
TopK	inactive	4	0.55	0.96	1.64	$< 10^{-6}$	0.4	1.37	2.07	0.432	0.745

F. Real-Model Pilot Details

Frozen choices. All of the following were fixed from calibration documents only and frozen, with a hash recorded, before any test document was read:

- the calibration feature pool (firing density in $[10^{-4}, 10^{-2}]$);
- the 64 sampled features and the pilot subsample of 8;
- per-feature doses ($q_{50}, q_{90}, q_{99}, 2q_{99}$ of positive calibration activations; $a_{\max}$ recorded but unused);
- norm budgets ($\{0.025, 0.05, 0.1, 0.2\}$ times the median centred residual norm).

Changes before and during the test run. A timing smoke on calibration documents projected an upper bound that exceeded the 30-minute budget, because its formula counted model loading and scanning once per group. Before the test run, we kept the pre-registered pilot size ($N = 8$ documents per feature and target type) after estimating the cost per quality group from the smoke’s own rows. Two rules were committed to the summary code while the test run was executing and before any test row existed on disk. They are *in-run amendments*, not pre-registered:

- gate readouts need at least 10 documents;
- a derived target-gain metric is reported.

Table 8. Exploratory sparser ReLU settings (added after development seeds).

Act.	Target	α	Rescaled decoder			LN correction				LN + repair	
			orig. act.	new	norm	tgt err	orig. act.	new	norm	tgt err	total
$b=-0.5$	active	0.25	0.562	0.23	1.5	$<10^{-6}$	$<10^{-6}$	0.244	1.86	$<10^{-6}$	$<10^{-6}$
$b=-0.5$	active	1	0.561	0.507	1.5	$<10^{-6}$	$<10^{-6}$	0.601	1.86	0.403	0.216
$b=-0.5$	active	4	0.558	0.892	1.5	$<10^{-6}$	$<10^{-6}$	1.16	1.86	0.515	0.269
$b=-0.5$	inactive	0.25	1.09	0.689	4.71	$<10^{-6}$	$<10^{-6}$	1.04	5.42	0.495	0.884
$b=-0.5$	inactive	1	0.498	0.77	2.35	$<10^{-6}$	$<10^{-6}$	0.92	2.79	0.67	0.24
$b=-0.5$	inactive	4	0.343	1.09	1.78	$<10^{-6}$	$<10^{-6}$	1.29	2.13	0.62	0.234
$b=-0.75$	active	0.25	0.137	0.0905	1.34	$<10^{-6}$	$<10^{-6}$	0.0693	1.38	$<10^{-6}$	$<10^{-6}$
$b=-0.75$	active	1	0.137	0.391	1.34	$<10^{-6}$	$<10^{-6}$	0.351	1.38	1.9×10^{-4}	0.0976
$b=-0.75$	active	4	0.137	0.789	1.34	$<10^{-6}$	$<10^{-6}$	0.828	1.38	0.484	0.18
$b=-0.75$	inactive	0.25	0.502	0.616	6.67	$<10^{-6}$	$<10^{-6}$	0.68	6.8	$<10^{-6}$	0.408
$b=-0.75$	inactive	1	0.212	0.628	2.87	$<10^{-6}$	$<10^{-6}$	0.689	2.94	0.545	0.493
$b=-0.75$	inactive	4	0.124	1.04	1.94	$<10^{-6}$	$<10^{-6}$	1.07	2.02	0.635	0.128

Table 9. Pre-registered dense ReLU setting (20 instances; medians). Drift is relative to α . “orig. act.” is D_{orig} and “new” is D_{new} (Definition 3.1). The last two columns are target error and total drift after repair. Other activations are in Table 7.

Act.	Target	α	Rescaled decoder			LN correction				LN + repair	
			orig. act.	new	norm	tgt err	orig. act.	new	norm	tgt err	total
ReLU	active	0.25	0.827	0.207	1.57	0.00955	0.0509	0.874	3.17	0.272	0.288
ReLU	active	1	0.786	0.51	1.57	0.00955	0.0509	1.73	3.17	0.501	0.301
ReLU	active	4	0.765	0.815	1.57	0.00955	0.043	2.43	3.17	0.558	0.338
ReLU	inactive	0.25	1.69	0.798	4.06	0.115	0.218	3.14	7.63	0.84	0.725
ReLU	inactive	1	0.862	0.803	2.26	0.0792	0.127	2.41	4.12	0.746	0.343
ReLU	inactive	4	0.603	1.07	1.82	0.0674	0.0931	2.38	3.51	0.653	0.307

Gate readouts (descriptive). The earlier gate conditions G1–G3 were read at q_{99} with document-cluster intervals, and all three fell on the continue side:

- G1: rescaled originally-active drift is above 0.1α . As run, it pools matched (active) and unmatched (inactive) rows; the split is in Section 5.2;
- G2: newly activated drift is below originally-active drift;
- G3: the correction lowers total drift.

The pilot configuration declares these readouts descriptive, not decisions.

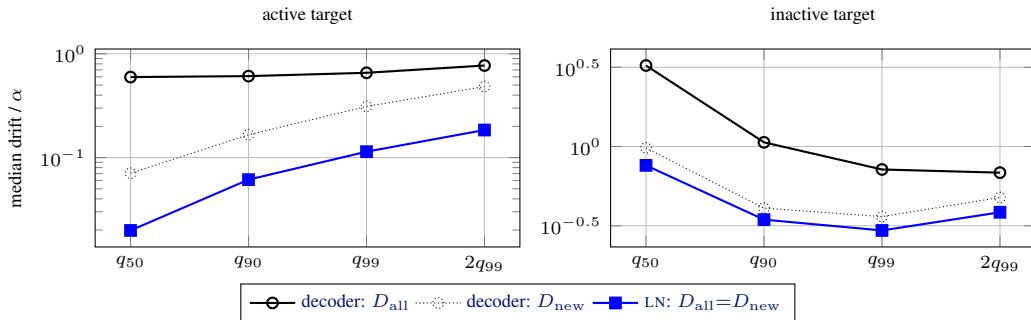


Figure 2. Pilot: drift versus calibration dose (target-matched; medians over test documents). The LN correction has no originally-active drift, so all of its drift is newly activated.

Table 10. Synthetic external proxy (20 instances; medians). “Err” is $\|\Delta z - \alpha e_j\|/\alpha$ measured with the SAE encoder (internal) or with A^+ (external). The last column counts instances where the LN correction beats the decoder externally.

Case	Method	Internal err	External err	Norm	LN better ext.
encoder = A^+	decoder	0.221	0.221	1	
	decoder_rescaled	0.222	0.222	1.04	
	encoder_grad	0.283	0.283	0.88	
	jacobian_ln	$< 10^{-6}$	$< 10^{-6}$	1	20/20
decoder = A	decoder	0.242	$< 10^{-6}$	1	
	decoder_rescaled	0.226	0.0708	0.997	
	encoder_grad	0.362	0.33	0.846	
	jacobian_ln	$< 10^{-6}$	0.239	0.964	0/20
both noisy	decoder	0.313	0.223	1	
	decoder_rescaled	0.306	0.241	1.06	
	encoder_grad	0.367	0.334	0.845	
	jacobian_ln	$< 10^{-6}$	0.223	0.986	11/20

G. P3-REAL-01R Protocol

Changes from the earlier design.

- The unit is the document, not the feature.
- $a_{\max}$ is replaced by calibration-only quantiles.
- Features and doses are frozen before any test window is encoded.
- Equal-norm and target-matched comparisons are reported separately.
- Infeasible rows are kept.

Gates. The original gate text is preserved:

- G1: “the rescaled decoder’s protected leakage is $< 0.1\alpha$ ”;
- G2: “crossing leakage $\geq$ protected leakage for the rescaled decoder”;
- G3: “the correction’s total leakage $\geq$ the rescaled decoder’s in $\geq 50\%$ of features”.

The condition used to be the median over test features at $\alpha \geq a_{\max}$. It is now read at q_{99} , per document, with 95% document-cluster bootstrap intervals, and each readout is one of stop side, continue side, or inconclusive. A large crossing share ends the method claim but not the measurement study.

Stages. Six command-line stages record their status (ran, blocked, not run) with time and peak memory: environment check, contract, calibration, timing smoke on calibration documents only, run, and summary. All six stages ran for the pilot (Section F); the full run over all 64 calibration features is NOT RUN.

H. Lexical Pilots: Protocols and Operational Records

Closed branch: wedding lexicon (P3-REAL-02-LX). The ActAdd keyword list (Turner et al., 2024) was frozen with the protocol below. Selecting the feature and the DiffMean direction requires at least 20 calibration windows that contain a keyword. With the frozen cap of 8 windows per article, 5 qualified and the stage returned BLOCKED (158 s). One amendment was committed before any test document was read: scan every calibration window, with a second BLOCKED ending the branch. All 1943 windows then gave 13 (630 s). The branch is closed: no test document was read and nothing was generated. This measures the lexicon’s support in WikiText, not a steering failure, and the minimum was not lowered.

Table 11. Contract execution: canonical TransformerLens/SAELens path versus the fast path (16 calibration windows; 96 admissible edits for K7). Tolerances were fixed before the float32 run; the float64 run is a diagnostic with the same tolerances.

Check	Quantity	Tol.	float32 (v1)	float64 (v2)
K0 weights/config identity	bitwise	—	✓	✓
K1 tokenizer (16 windows)	equal	—	16/16	16/16
K2 activation, $x = h - \bar{h}$	rel. ℓ_2	10^{-4}	4.2×10^{-6}	1.0×10^{-14}
uncentred h (neg. control)	rel. ℓ_2	—	0.0319	0.0319
K3 SAE features	rel. ℓ_2	10^{-3}	7.7×10^{-6}	1.6×10^{-14}
K4 L0 / FVE (canonical)	band	[30,120] / 0.7	72.4 / 0.827	72.4 / 0.827
uncentred input	—	—	176 / 0.003	176 / 0.003
K5 no-op hooks	max $ \Delta $	0	0	0
K6 common logit shift	max $ c_t $	—	279	279
residual after shift	max	10^{-3}	4.3×10^{-4}	7.4×10^{-13}
probabilities	max $ \Delta p $	10^{-5}	2.8×10^{-5}	5.1×10^{-14}
KL	max	10^{-6}	7.5×10^{-7}	1.5×10^{-15}
K7 post-edit feature change	rel. ℓ_2	10^{-3}	7.6×10^{-5}	1.9×10^{-13}
post-edit probabilities	max $ \Delta p $	10^{-5}	2.8×10^{-5}	5.3×10^{-14}
post-edit KL	max	10^{-6}	7.7×10^{-7}	1.5×10^{-15}
K9 mean-only edit, model KL	max	10^{-8}	6.0×10^{-7}	1.3×10^{-15}
SAE drift / ρ , raw off-slice $E(x+\delta)$	median	—	125	125
SAE drift / ρ , canonical $E(P(h+\delta))$	max	—	not run	3.8×10^{-15}
K10 no-processing path, KL	max	10^{-6}	not run	1.9×10^{-15}
Verdict	FAIL (blocked) PASS (float64)			

P3-REAL-02-SPACE protocol. A new configuration was committed before any corpus count, feature forward or generation. It differs from the wedding protocol (after its amendment) only in the following:

- the topic, set by instruction rather than by searching topics: the 18-word space list of PPLM (Dathathri et al., 2020), from the official repository at a pinned commit (Apache 2.0). Only the lexical target is used; the PPLM method is not reproduced;
- explicit matching rules. Text is lowercased, and a word is a maximal run of ASCII letters. Entries match whole words exactly, with no substrings, stemming or inflection. A word cut by a window or prompt edge never matches. Counts use decoded text, not token strings;
- a text-only support check that runs first: 58 of 1943 calibration windows contain a list word (minimum 20). They come from 22 of 60 articles, with at most 9 in one article. Windows are not independent documents, and 28 of the positive windows contain the polysemous “star”;
- one confirmatory comparison, fixed in advance: the per-document difference in any-hit between LN and the decoder. Everything else is descriptive.

Unchanged from the wedding protocol:

- calibration on WikiText-103 validation articles; the feature is the largest activation difference between windows with and without list words, and DiffMean is the difference of mean centred hook states (the lexicon therefore also selects the directions);
- each actor’s norm budget is the largest of $\{0.05, 0.1, 0.2, 0.4, 0.8\}$ times the median centred norm whose mean teacher-forced $\text{KL}(P_{\text{base}} \| P_{\text{edit}})$ on 32 calibration prompts is at most $\kappa = 0.1$ nats. A method with no such budget is NOT RUN, not substituted;
- test articles in a fixed seeded order, one keyword-free prompt each (BOS plus 31 tokens), up to 32 articles by a cost rule; every attempted article is recorded;
- greedy decoding of 32 tokens with the edit at every position and no KV cache;

Table 12. Contract fields, sources and status after execution. “Verified” means read from the pinned source file or code; “executed” means checked by running both paths (Table 11).

Field	Value	Status
Model	GPT-2 small, pinned HF revision, MIT (card)	verified
SAE	residual SAE, <code>blocks.8.hook_resid_pre</code> , pinned revision, MIT (card)	verified; weights bitwise equal to the SAE lens load
Width, dtype, context	$d = 768$, $m = 24576$, float32 weights, 128	verified (config, header)
Activation, input bias, normalization	ReLU; $x - b_{\text{dec}}$; none	library defaults, executed (K0, K3)
Coordinates	HF hidden state minus per-token mean	executed (K2, K4)
Tensors	$W_{\text{enc}} [768 \times 24576]$, $W_{\text{dec}} [24576 \times 768]$, biases, F32	verified (header)
Tokenizer, BOS, positions	GPT-2 BPE, BOS prepended, absolute positions	executed (K1)
Processing options	<code>fold_ln</code> , <code>center_writing_weights</code> , <code>center_unembed</code> , <code>fold_value_biases</code> on (explicit)	executed; no-processing path agrees (K10)
Model-visible edits	classes $\delta + \text{span}(\mathbf{1})$, represented in $\text{range}(P)$, $P = I - \mathbf{1}\mathbf{1}^\top/d$	proved for GPT-2 (Proposition 4.1); executed (K9)
SAE readout of an edit	canonical $E(P(h + \delta)) - E(P_h)$	readout closure (Section 4)
Data	WikiText-103 raw (CC BY-SA 3.0 / GFDL on card)	verified (card)
Libraries	TransformerLens 2.18.0, SAE lens 6.51.3, torch 2.14.0 (CPU)	pinned

Table 13. Pilot row status by method (target-matched mode, all four doses, active and inactive targets). “Raw-only”: rows that exist only under the raw off-slice readout. “Docs”: largest number of test documents in one dose and target-type cell.

Method (target-matched, 4 doses)	rows	matched	unmatched	infeasible	raw-only	docs
decoder	512	512	0	0	0	42
encoder row	512	512	0	0	0	42
LN ($J_E P$)	512	512	0	0	0	42
random	512	240	0	272	0	42
rescaled (weight-only)	512	0	512	0	0	42
plain decoder (αD_j)	512	0	512	0	0	42
mean-only (raw readout)	512	0	0	0	512	42

- the mean-only control at the largest budget, outside every ranking;
- random directions, SAE-TS and FGAA are NOT RUN.

I. Reproducibility, Compute and Claim Map

- *Toy suite*. It runs in 120 s or less in the standard library from a single configuration. Tables, figure data and every number in the text are regenerated by one script.
- *Real-model runs*. They used pinned revisions and a pinned environment (package list in the results), in float64 on two CPU threads. Wall-clock times:
 - contract execution: 161 s in float32 and 391 s in float64;
 - adapter contract stage: 71 s; calibration: 330 s; timing smoke: 95 s; pilot run: 1043 s; summary: 34 s;
 - readout closure: 843 s wall (1481 CPU-seconds summed over the two threads);
 - matched-denominator re-aggregation: about 5 s (standard library, one thread);
 - wedding lexicon: calibration 158 s and 630 s (both blocked);
 - space lexicon: text-only support count 11 s; calibration 961 s; run 1334 s (32 articles, 192 rows); summary about 1 s.

Peak memory was below 9 GB. No GPU was used. Model, SAE and data files are pinned by revision and sha256; the only later external file is the 142-byte PPLM word list (with its license).

- *Claim map*. A claim-to-evidence table (`claims.csv`) links each statement to its files, experiment identifiers, assumptions and status.

Table 14. Pilot, target-matched at q_{99} : medians with 95% document-cluster bootstrap intervals (drift part). k : k infeasible rows (kept, not dropped). Quality and norm are in Table 15.

Target	Method	Target err	D_{orig}/α	D_{new}/α	D_{all}/α
active	decoder	$< 10^{-7}$	0.583 [0.452, 0.6]	0.311 [0.236, 0.342]	0.656 [0.55, 0.704]
	rescaled (weight-only)	$< 10^{-7}$	0.583 [0.452, 0.6]	0.311 [0.236, 0.342]	0.656 [0.55, 0.704]
	encoder row	$< 10^{-7}$	0.681 [0.608, 0.725]	0.366 [0.314, 0.45]	0.778 [0.691, 0.935]
	LN ($J_E P$)	$< 10^{-7}$	4.7×10^{-15} [3.6×10^{-15} , 5.7×10^{-15}]	0.114 [0.0976, 0.135]	0.114 [0.0976, 0.135]
	random ³³	$< 10^{-7}$	7.33 [4.32, 10.9]	108 [46.8, 237]	108 [46.9, 238]
inactive	decoder	$< 10^{-7}$	0.583 [0.485, 0.685]	0.361 [0.303, 0.416]	0.716 [0.59, 0.819]
	rescaled (weight-only)	0.727 [0.629, 0.856]	0.342 [0.289, 0.371]	0.112 [0.0993, 0.125]	0.37 [0.322, 0.394]
	encoder row	$< 10^{-7}$	0.943 [0.707, 1.2]	0.943 [0.784, 1.07]	1.36 [1.13, 1.53]
	LN ($J_E P$)	$< 10^{-7}$	5.0×10^{-15} [4.0×10^{-15} , 6.5×10^{-15}]	0.295 [0.27, 0.342]	0.295 [0.27, 0.342]
	random ³⁵	$< 10^{-7}$	17.4 [7.69, 32]	347 [222, 604]	348 [222, 604]

Table 15. Pilot, target-matched at q_{99} (quality part): KL over the 16 positions after the edit, in units of 10^{-3} nats, and edit norm, with 95% document-cluster bootstrap intervals.

Target	Method	KL $\times 10^3$	$\ \delta\ $
active	decoder	5.32 [0.523, 14.8]	21.6 [12.8, 30.8]
	rescaled (weight-only)	5.32 [0.523, 14.8]	21.6 [12.8, 30.8]
	encoder row	1.03 [0.177, 2.72]	11.8 [8.17, 17]
	LN ($J_E P$)	0.96 [0.14, 3.19]	12.7 [9.47, 18.5]
	random ³³	691 [66.2, 950]	438 [271, 875]
inactive	decoder	3.41 [1.37, 9.69]	36.9 [29, 41.5]
	rescaled (weight-only)	0.65 [0.431, 5.91]	22.5 [12.3, 26.5]
	encoder row	1.38 [0.485, 2.49]	21 [17.2, 23.7]
	LN ($J_E P$)	1.16 [0.393, 3.19]	22.5 [19.4, 25.2]
	random ³⁵	327 [137, 980]	1174 [594, 1837]

- *Study status.* METHOD UTILITY NOT SUPPORTED IN THIS PILOT; STUDY SCOPE FROZEN. This is the status of this study, not a verdict on SAE steering in general. Reopening the behavioral question would need a behavioral label defined independently of the direction selection, data separated from calibration, a design with adequate precision, and a question posed against the closest prior work.
- *Supplement.* The anonymous supplement contains the code, configurations, tests, raw result files, the claim map (claims.csv) and the space word list with its license. Model, SAE and data files are not included; they are pinned by revision and sha256.

Table 16. Pilot: per-document paired differences, LN minus decoder (target-matched), with 95% document-cluster bootstrap intervals. ΔKL is in units of 10^{-3} nats; the originally-active drift of the LN edit is zero by construction, so its paired difference equals minus the decoder’s D_{orig} in Table 14.

Target	Dose	n_{doc}	ΔD_{all}	ΔD_{new}	ΔKL
active	q50	42	-0.542 [-0.586, -0.455]	-0.0337 [-0.0499, -0.0238]	-0.0114 [-0.0371, -0.00643]
	q90	42	-0.52 [-0.577, -0.438]	-0.0927 [-0.115, -0.0686]	-0.229 [-0.715, -0.0442]
	q99	42	-0.503 [-0.584, -0.421]	-0.17 [-0.202, -0.142]	-3.77 [-11.7, -0.248]
	2xq99	42	-0.518 [-0.596, -0.421]	-0.282 [-0.304, -0.216]	-12.1 [-50.2, -0.883]
inactive	q50	40	-2.58 [-2.97, -2.24]	-0.119 [-0.259, 0.0136]	-0.397 [-0.634, -0.102]
	q90	40	-0.741 [-0.925, -0.597]	-0.0729 [-0.0909, 0.0173]	-0.733 [-1.49, -0.316]
	q99	40	-0.43 [-0.517, -0.35]	-0.0557 [-0.098, -0.00262]	-2.34 [-6.51, -0.868]
	2xq99	40	-0.294 [-0.348, -0.25]	-0.0467 [-0.133, -0.0127]	-5.89 [-21.2, -2.05]

Table 17. Pilot, equal norm ($\rho = 0.1 \times$ median residual norm). “Target gain” is the target-feature change per unit edit norm, a metric added as an in-run amendment (Section 5.2). Mean-only rows are omitted (raw off-slice readout).

Target	Method	target gain	D_{all}/ρ	KL (nats)
active	decoder	0.895 [0.847, 0.946]	0.517 [0.473, 0.53]	0.00073 [0.00042, 0.00159]
	encoder row	1.5 [1.47, 1.72]	1.16 [0.967, 1.26]	0.00080 [0.00051, 0.00119]
	LN ($J_E P$)	1.35 [1.29, 1.55]	0.142 [0.126, 0.152]	0.00065 [0.00036, 0.00084]
	random	0.0383 [0.029, 0.0483]	0.341 [0.319, 0.361]	0.00046 [0.00038, 0.00048]
inactive	decoder	0 [0, 0.0494]	0.316 [0.303, 0.335]	0.00052 [0.00024, 0.00063]
	encoder row	0.433 [0.303, 0.476]	0.859 [0.726, 0.984]	0.00052 [0.00032, 0.00068]
	LN ($J_E P$)	0.304 [0.174, 0.386]	0.125 [0.101, 0.141]	0.00044 [0.00029, 0.00063]
	random	0 [0, 0]	0.348 [0.32, 0.38]	0.00027 [0.00023, 0.00059]

Table 18. Space-lexicon pilot, paired per article (mean differences with 95% document-bootstrap intervals; only the first row is confirmatory). “a only / b only”: discordant articles for the primary pair.

Pair ($a - b$)	n	Δ any hit (pp)	a only / b only	Δ test KL	Δ NLL
LN – decoder (primary)	32	0 [0, 0]	0 / 0	-0.00942 [-0.0115, -0.00735]	0.0395 [-0.0595, 0.151]
LN – DiffMean	32	-3.12 [-12.6, 6.25]	–	-0.0283 [-0.0331, -0.0235]	-0.0449 [-0.21, 0.12]
LN – encoder row	32	0 [0, 0]	–	-0.00869 [-0.0105, -0.00692]	0.0225 [-0.0706, 0.128]
decoder – DiffMean	32	-3.12 [-12.6, 6.25]	–	-0.0189 [-0.0242, -0.0136]	-0.0843 [-0.233, 0.0529]
decoder – no edit	32	0 [0, 0]	–	0.0295 [0.0271, 0.0322]	-0.00219 [-0.107, 0.098]
encoder row – no edit	32	0 [0, 0]	–	0.0288 [0.0264, 0.0312]	0.0148 [-0.0784, 0.119]
LN – no edit	32	0 [0, 0]	–	0.0201 [0.0191, 0.0211]	0.0373 [-0.0865, 0.175]
DiffMean – no edit	32	3.12 [-6.25, 12.6]	–	0.0485 [0.0432, 0.0537]	0.0822 [-0.0629, 0.226]